Devless

AI-Powered Code Generation from GitHub Issues

An army of AI workers at your service

⚠️ **This is a Proof of Concept** ⚠️

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The Problem

Problem to solve

- Create issues → get working code
- Review-ready PRs
- Respond to PR comments, iterate

Ideal scenario

> Open issue → AI implements → PR opens → Review → Merge

Personal Motivation (gaining experience with...)

- Kiro Spec coding mode
- Deeper experience in Architectures built around AI
- security + LLM
- finding balance between LLM-based vs deterministic impl
- SKILLS.md approach

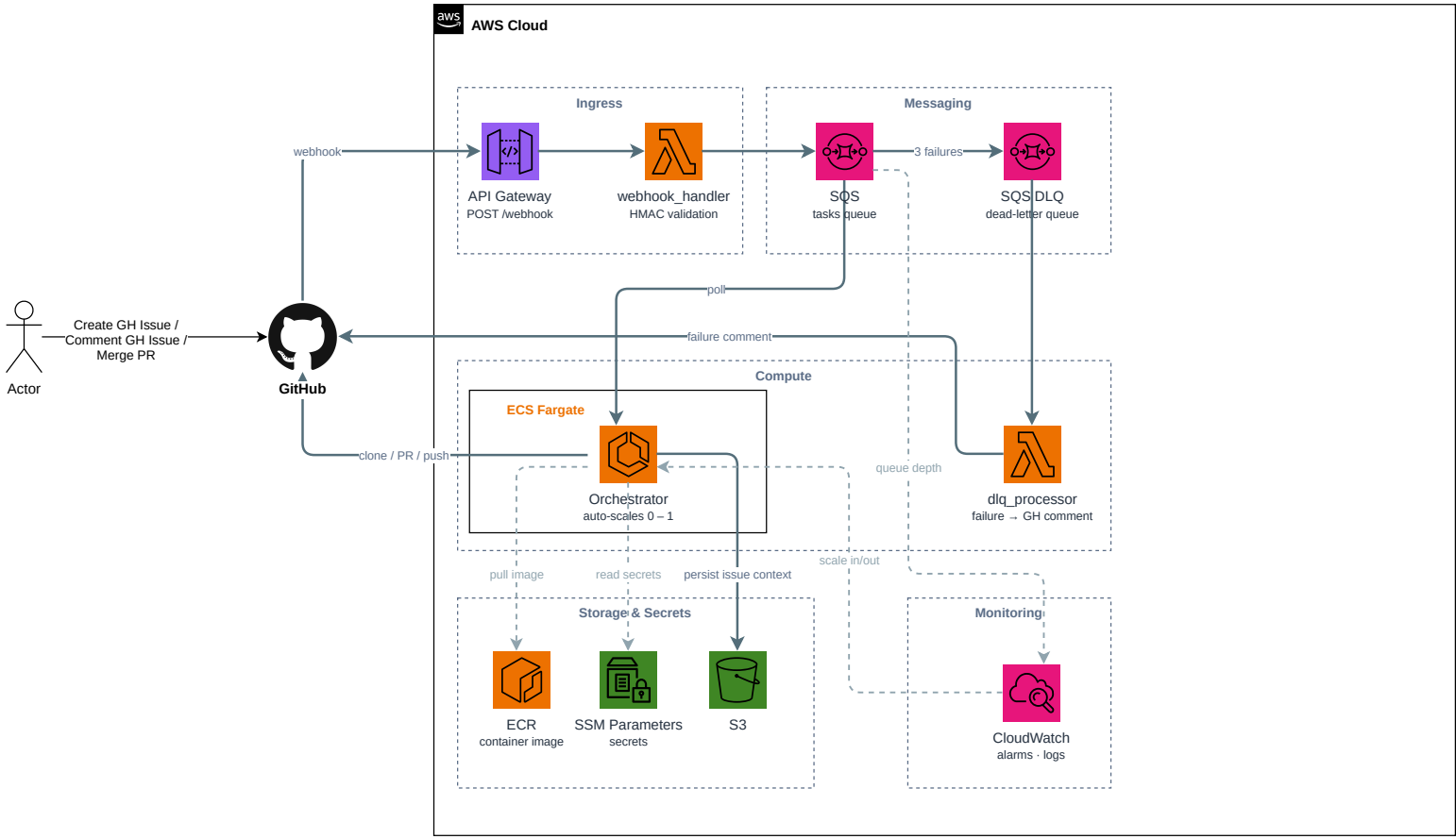
Available Tools

Tool	Status	Notes
GitHub Copilot Agent	✅ Tested	Works well, zero setup
Devin	❓ Not tested	$500/mo
SWE-agent	❓ Not tested	DIY setup
Aider	✅ Used	CLI, no GitHub integration

Motivation for POC: AWS/Bedrock self hosted solution. How hard is it?

Solution: Devless

Issue → Webhook → SQS → Orchestrator → Aider → PR



Two Orchestrator Approaches

1. State Machine (Original)

- Python code controls flow
- Explicit states: `pending → cloned → agent_loop → pr_created`
- ~800 lines of code

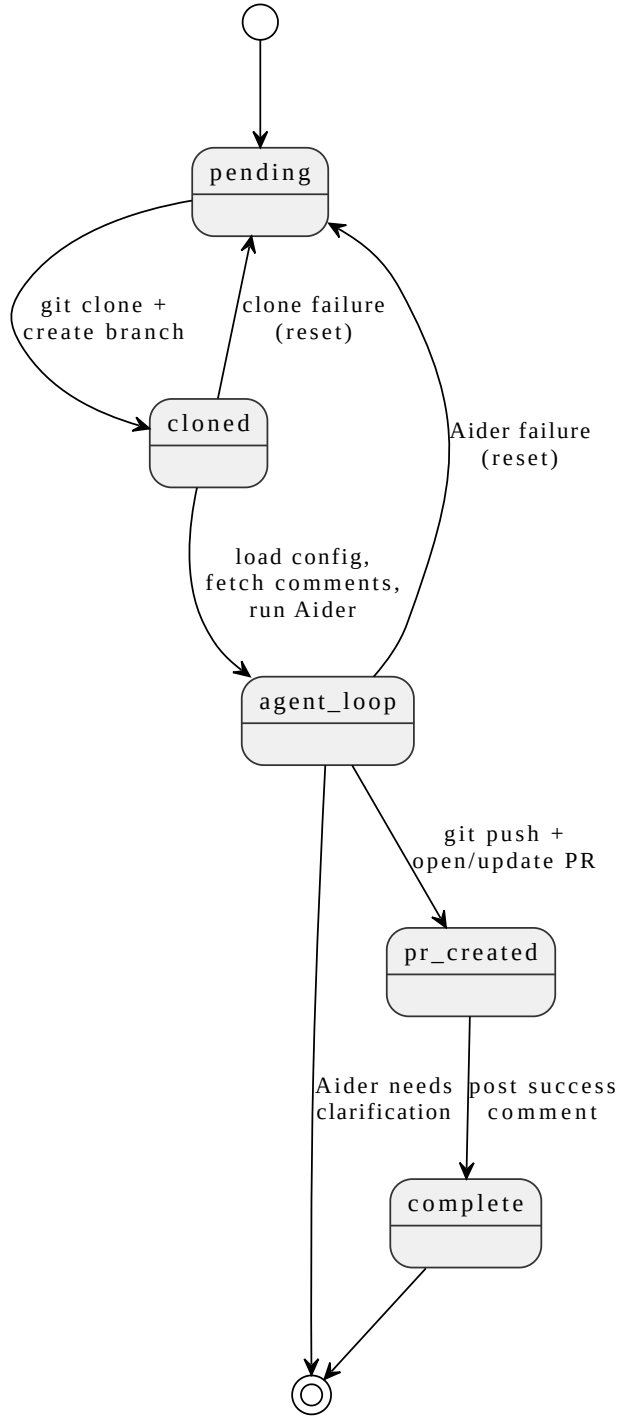
2. Agent-based (Strands)

- LLM decides flow via SKILL.md files
- Tools for git/GitHub/code operations
- ~400 lines of code

State Machine Approach

Pros: Deterministic, easy to debug

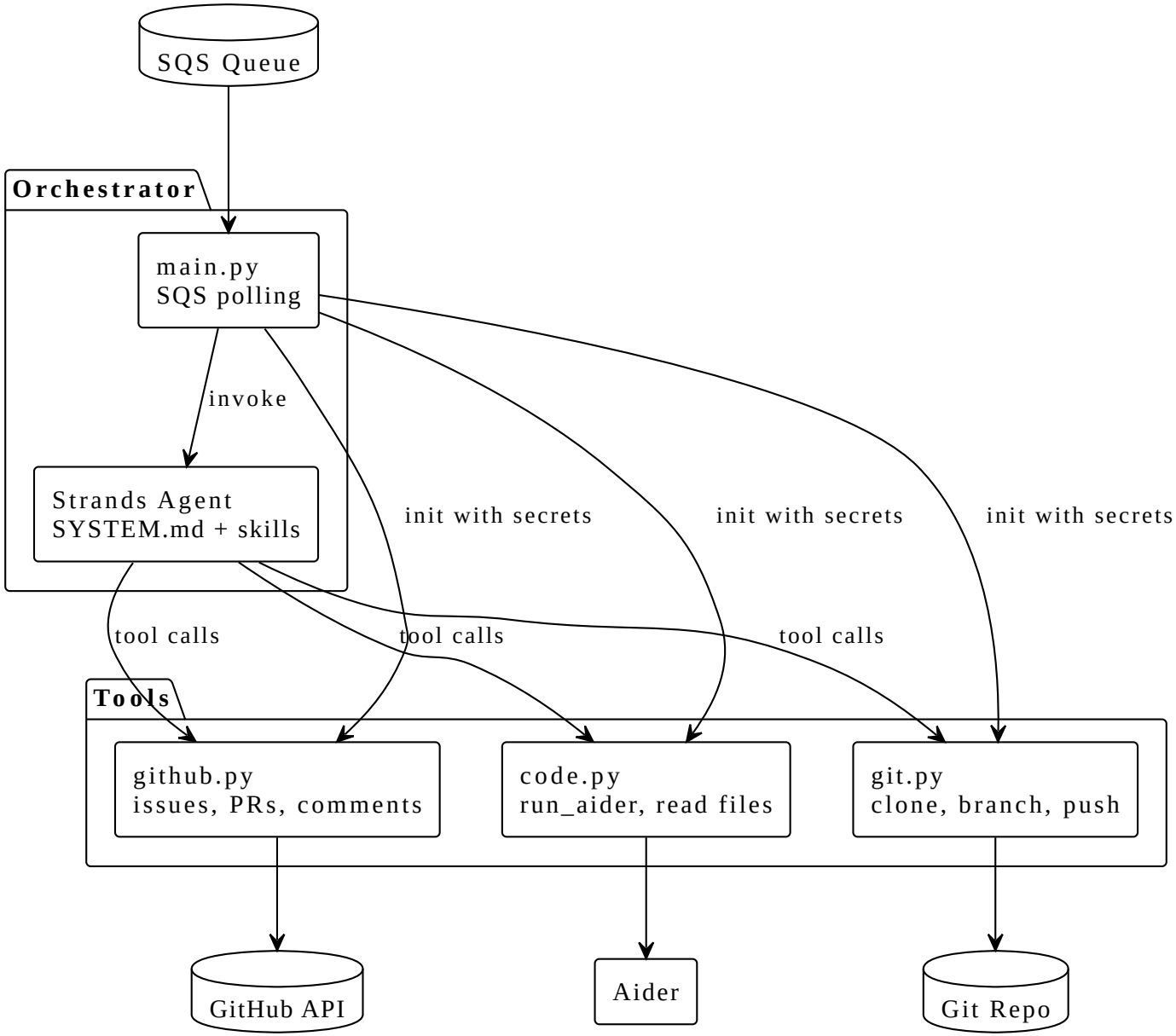
Cons: Rigid, code changes for new workflows



Agent-based Approach

Pros: Flexible, edit SKILL.md for changes

Cons: LLM behavior may vary



Comparison: Two Approaches

Aspect	State Machine	Agent-based
Control flow	Hardcoded Python	LLM via SKILL.md
Lines of code	~800	~400
Adding workflows	Code changes	Edit SKILL.md
Error handling	try/except	Agent decides
Predictability	✅ Deterministic	⚠️ May vary
Flexibility	⚠️ Rigid	✅ Adapts
Secrets	Hidden	Hidden in tools

Security: Secrets Handling

Both approaches hide secrets from LLM

State Machine: Orchestrator holds token, never passes to Aider

Agent-based: Tools initialized with token at startup

- LLM calls `create_pr(title, body)`
- Tool internally uses token
- Token never in LLM context

vs GitHub Copilot Agent

Aspect	Devless	Copilot Agent
Setup	Terraform deploy	✅ Zero
Cost	~$20/mo	GitHub sub
LLM choice	Bedrock, OpenAI	GitHub only
Customization	Full control	Limited
Self-hosted	✅ Yes	❌ No
Data residency	Your AWS	GitHub
Maintenance	You	GitHub

Demo

Repo: github.com/typekpb/devless-demo

1. Open issue describing feature
2. Devless posts status comment
3. PR opens with implementation
4. Review, comment, iterate
5. Merge

Target Repo Setup

`.devless/config.yml`:

```
model: bedrock/eu.anthropic.claude-sonnet-4-20250514-v1:0
test_command: npm test
```

Webhook events: Issues, Issue comments, PR reviews

That's it. No agents to install.

Infrastructure

```
terraform/
├── modules/
│   ├── api_gateway/   # Webhook endpoint
│   ├── lambda/        # HMAC validation
│   ├── sqs/           # Task queue
│   ├── ecs/           # Orchestrator
│   └── iam/           # Roles
```

One `terraform apply` deploys everything.

Development Cost & Timeline

Item	Value
Kiro subscription	$20
AWS sandbox	$0
Kiro credits used	~900 (free 50 ran out fast)

Lessons Learned

What worked ✅

- Aider — battle-tested
- GitHub App auth — no PATs
- Bedrock — zero config with IAM

Key insight

- State machines feel safe but become rigid.
- Agent + tools = less code, same control.

Lessons Learned

Model matters ⚠️

- Haiku got stuck on simple TypeScript error (unused import)
- Kept adding `import React` instead of removing it
- Sonnet understood the fix immediately

Recommendation: Start with agent-based, use Sonnet or better.

Coding Styles Comparison

Style	Feedback Loop	Best For	Target Users
Vibe coding	✅ Instant	Quick prototypes	Developers
Specs (Kiro)	⚠️ Upfront planning	New projects only	Architects, PMs
GitHub Issues	⚠️ Slower	Existing projects	Non-devs, PMs

Coding Styles Comparison

Specs approach

- Nice for planning ahead
- But feels like waterfall 🚫
- Only works for greenfield projects

GitHub Issues approach

- Slower feedback (wait for PR)
- **Enables non-developers** to request features
- **Enhancement idea:** Stream Aider output to issue comments

Q&A

Devless: github.com/typekpb/devless

Demo: github.com/typekpb/devless-demo